Mock exam 1

3 hours · 100 points

About this mock exam. This mock exam was written by an AI model, from written instructions on how the exam is structured and what it should examine. It is based on the ordinary exam of 2025, adjusted for length and difficulty: that exam was a 12-hour take-home exam with every aid allowed, while this year's exam is three hours at the department, without aids.

Some of it is on lectures still to come. Part A is on lecture 9 (network effects in green technology adoption, week 46), essay C.1 on lecture 10 (the energy efficiency gap, week 47) and essay C.3 on lecture 13 (energy and geopolitics, week 50). Part B (lectures 2, 4 and 6) and essay C.2 (lecture 5) are on material you have already had.

The solutions, with the marking guide for every question, are in a separate PDF. Try the exam first.

Instructions. The exam has three parts. Each part counts for the points given, and the points sum to 100. Answer all questions. Write your answers on the computer; derivations may be typed in Word or LaTeX.

No aids are allowed. You are not expected to remember numbers from the readings or to cite them: an answer is complete when it gives the mechanism or the argument the question asks for.

Within a model part, a question that states a result can be answered without the questions before it: use the stated result even if you did not derive it.

The essays in part C should each be at most 200 words.

Part		Points
A	Electric vehicles and charging stations	35
B	Battery storage in a market with wind and solar	35
C	Essays	30

Electric vehicles and charging stations

Electric vehicles (EVs) need charging stations, and charging stations need EVs. This part studies the feedback between the two with a stylised version of the model of Li et al. (2017).

At the start of period t there are Q_t EVs on the road and N_t public charging stations. During period t, consumers buy q_t new EVs at the price p_t, and a share $\delta \in (0, 1)$ of the existing stock survives into the next period. Station owners build stations in response to the stock of EVs. The model is

$$\ln q_t = \beta_1 \ln N_t + \beta_2 \ln p_t, \quad \beta_1 > 0,\ \beta_2 < 0, \quad \text{(A1)}$$

$$\ln N_t = \gamma_1 \ln Q_t, \quad \gamma_1 > 0, \quad \text{(A2)}$$

$$Q_{t+1} = q_t + \delta Q_t. \quad \text{(A3)}$$

A.1 4 points

Explain the difference between a direct and an indirect network effect, and why the relation between EVs and charging stations is an indirect one. Why might it be a reason for policy to intervene?

A.2 5 points

Interpret β_1, β_2 and γ_1. Then show that the stock of EVs evolves according to

$$Q_{t+1} = f(Q_t) \equiv Q_t^{\beta_1\gamma_1}\, p_t^{\beta_2} + \delta\, Q_t. \quad \text{(A4)}$$

A.3 8 points

Let the EV price be constant, $p_t = p$. Derive the steady state $Q^* > 0$. Show that the slope of the transition function at the steady state is

$$f'(Q^*) = \beta_1\gamma_1(1 - \delta) + \delta,$$

and use it to state the condition under which Q^* is stable. Draw the transition diagram for this case, and explain in words what the condition says about the feedback between cars and chargers. Does the model have a critical mass?

A.4 7 points

Assume the condition from A.3 holds. Show that the long-run elasticity of EV sales with respect to the price, in the steady state, is

$$\frac{d \ln q^*}{d \ln p} = \frac{\beta_2}{1 - \beta_1\gamma_1},$$

and compare it with the elasticity of sales in a single period, holding the stock of stations fixed. What does the comparison say about the effect of a permanent purchase subsidy? And what does a subsidy paid only for a few periods do to the path of Q_t and to the steady state?

A.5 5 points

Now suppose the feedback is strong: $\beta_1 \gamma_1 > 1$. Draw the transition diagram, and say which steady states are stable. Does the model now have a critical mass? What does your answer imply for a subsidy paid only for a few periods?

A.6 6 points

Li et al. (2017) find that the US federal tax credit for EV purchases could have bought considerably more EVs had the same money been spent on subsidising charging stations. Using the model, explain what makes a station subsidy more cost-effective than a purchase subsidy. Would you expect the conclusion to carry over to Denmark today, where EVs make up most new car sales? Does the paper's result show that subsidising charging stations is socially optimal?

Part B · Model part 35 points

Battery storage in a market with wind and solar

A firm owns a battery in a single price zone and takes the hourly wholesale prices λ_h, $h = 1, \dots, T$, as given. The battery has a power rating $\bar{s}$ (MW), the most it can charge or discharge in an hour, and a duration $\bar{\delta}$ (hours), so that it holds at most $\bar{\delta}\,\bar{s}$ MWh. Charging g_h^c MWh from the grid puts $\eta_c g_h^c$ MWh into the battery, and delivering g_h^d MWh to the grid takes g_h^d/η_d MWh out of it, with $\eta_c, \eta_d \in (0,1)$. Every MWh charged or discharged costs c^s in wear. The stored energy at the end of hour h is soc_h, and the year is cyclical: the battery must end the year with the energy it started with.

B.1 6 points

For a given power rating and duration, write down the firm's problem of choosing charging, discharging and stored energy in every hour to maximise its operating profit over the year. Explain each constraint in a line.

B.2 7 points

With θ_h the multiplier on the law of motion for stored energy, $\underline{\phi}_h^c, \overline{\phi}_h^c$ and $\underline{\phi}_h^d, \overline{\phi}_h^d$ those on the charging and discharging limits, and $\underline{\psi}_h, \overline{\psi}_h$ those on the energy limit, the first-order conditions are

$$g_h^c: \; -c^s - \lambda_h + \eta_c\,\theta_h + \underline{\phi}_h^c - \overline{\phi}_h^c = 0,$$

$$g_h^d: \; -c^s + \lambda_h - \theta_h/\eta_d + \underline{\phi}_h^d - \overline{\phi}_h^d = 0,$$

$$soc_h: \; -\theta_h + \theta_{h+1} + \underline{\psi}_h - \overline{\psi}_h = 0.$$

What does θ_h measure? Use the conditions to explain in which hours the battery charges, discharges and does neither, and why θ_h can change from one hour to the next only when a limit on stored energy binds.

B.3 6 points

The firm's battery is small, but suppose many batteries enter the zone. What do they do to the distribution of prices over the year, and what does that do to what the next battery can earn? Would a planner who chose the batteries' operation to minimise system cost operate them differently from price-taking firms?

B.4 6 points

Now let the firm also choose the power rating $\bar{s} \geq 0$ (multiplier $\underline{\beta}_s$) of a battery with a given duration $\bar{\delta}$, at an annual fixed cost F_s per MW. The first-order condition for $\bar{s}$ is

$$-F_s + \sum_h \left(\overline{\phi}_h^c + \overline{\phi}_h^d\right) + \bar{\delta} \sum_h \overline{\psi}_h + \underline{\beta}_s = 0.$$

Interpret the condition for a battery that is built. In which hours does a MW of battery earn its keep, and what decides how much it earns?

B.5 5 points

In the greenfield model of ESM section 7, the planner can build batteries of 2, 4 or 8 hours' duration, and the reference case builds only 8-hour batteries. Figure 1 shows the median DK2 price in each hour of the day, month by month, in that case. Use the figure and the condition from B.4 to explain why the longest battery is the one built. Which duration would you expect to be built if the price instead spiked for a single hour each evening and were flat otherwise?

B.6 5 points

Suppose the grid company charges the battery a volumetric network tariff τ per MWh it draws from the grid, as it does any consumer. How does the tariff change the battery's operation and its investment decision? Argue whether, and when, such a tariff is justified.

Essays

Answer each question in at most 200 words. An answer should be convincing, complete and concise; you do not need to cite sources or numbers.

C.1

10 points

Engineering studies often find that energy efficiency investments would save money and cut emissions at the same time. Explain why evaluations made after a programme has run tend to find smaller gains than the projections made before it, and what this implies for subsidising energy efficiency as climate policy. *at most 200 words*

C.2

10 points

Do heat pumps help integrate wind power? Discuss what a growing fleet of heat pumps does to the cost of the electricity they use and to the market value of wind power, and what making the heat pumps flexible changes. *at most 200 words*

C.3

10 points

The G7 capped the price at which Russian oil may be transported with Western shipping and insurance. Explain why a lower cap need not reduce Russia's oil profits once Russia can build a shadow fleet, and what this implies for the design of sanctions. *at most 200 words*

Figures

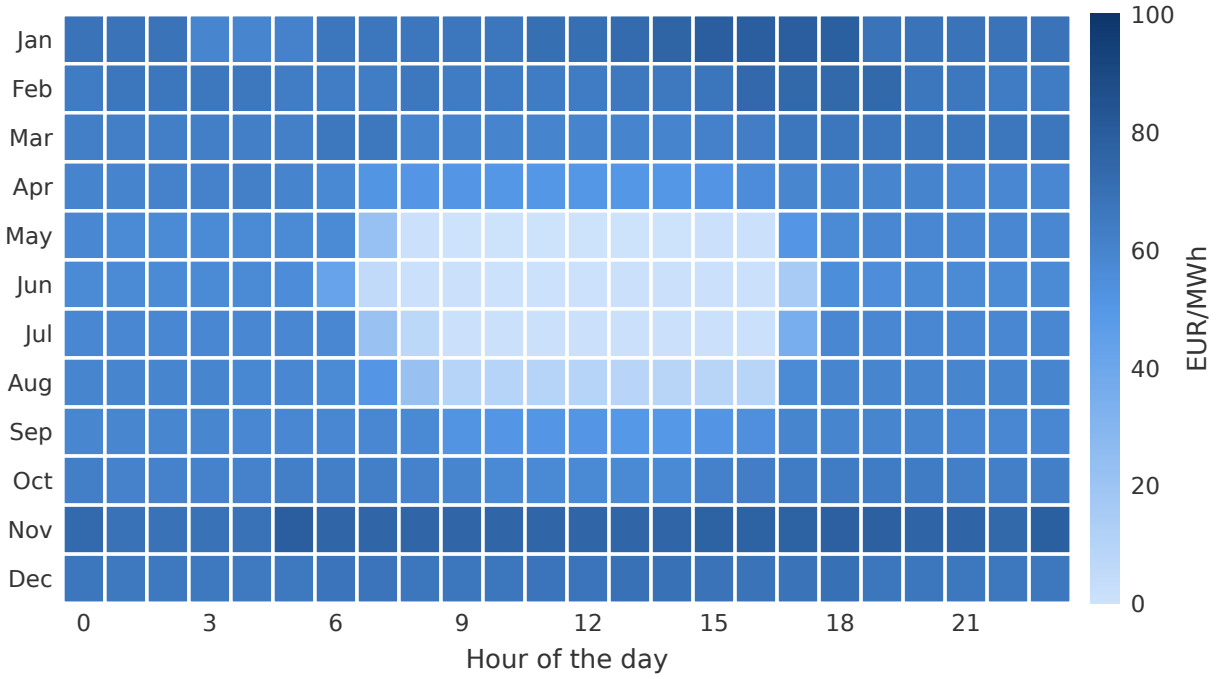

Figure 1. The Model Explorer, reference case of ESM section 7: the median DK2 price in each hour of the day, month by month (Operation › The price through the day, month by month). On half the days of the month the price at that hour was higher.